

CS 236: Deep Generative Models

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Stanford University

URL: deepgenerativemodels.github.io

Introduction

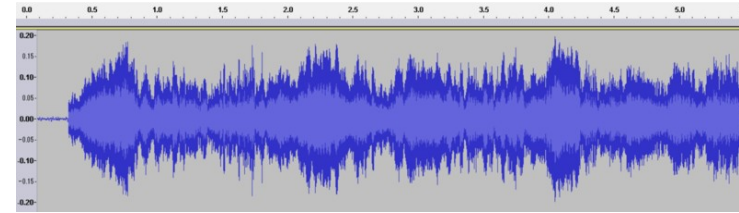
Challenge: understand complex, unstructured inputs



Computer
Vision



Natural Language
Processing

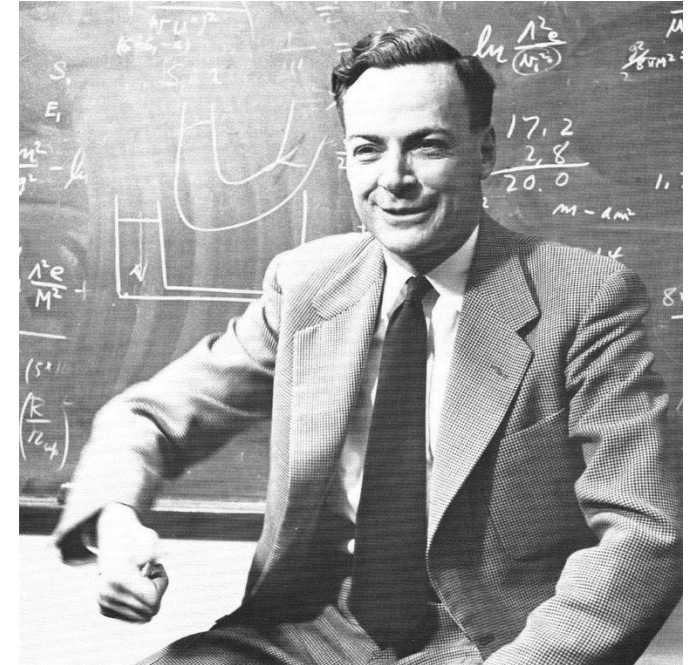


Computational
Speech



Robotic
s

Introduction



Richard Feynman: *"What I cannot create, I do not understand"*

Generative modeling: *"What I understand, I can **create**"*

Generative Modeling: Computer Graphics

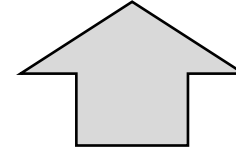
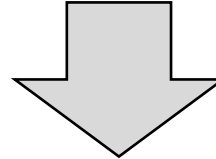
How to generate natural images with a computer?

High level
description

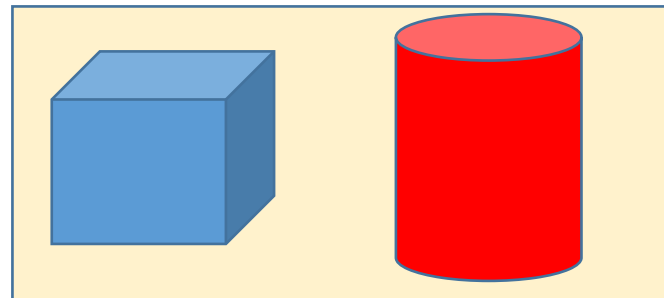
```
Cube(color=blue, position=(x,y,z), size=...)  
Cylinder(color=red, position=(x',y',z'), size=..)
```

Generation
(graphics)

Raw sensory
outputs



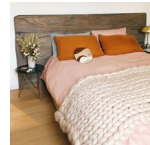
Inference (vision as
inverse graphics)



Many of our models will have **similar structure (generation + inference)**

Statistical Generative Models

Statistical generative models are **learned from data**



...



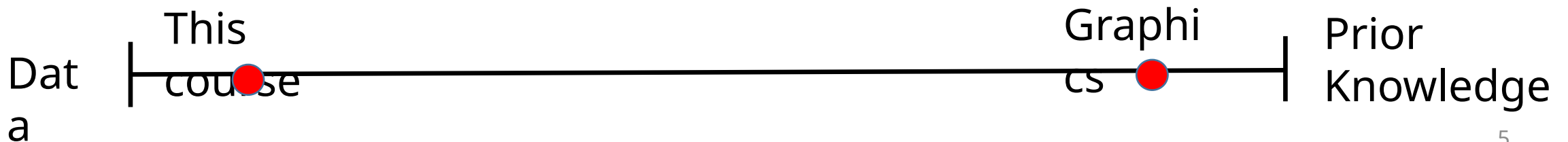
+



Data
(e.g., images of
bedrooms)

Prior Knowledge
(e.g., physics,
materials, ..)

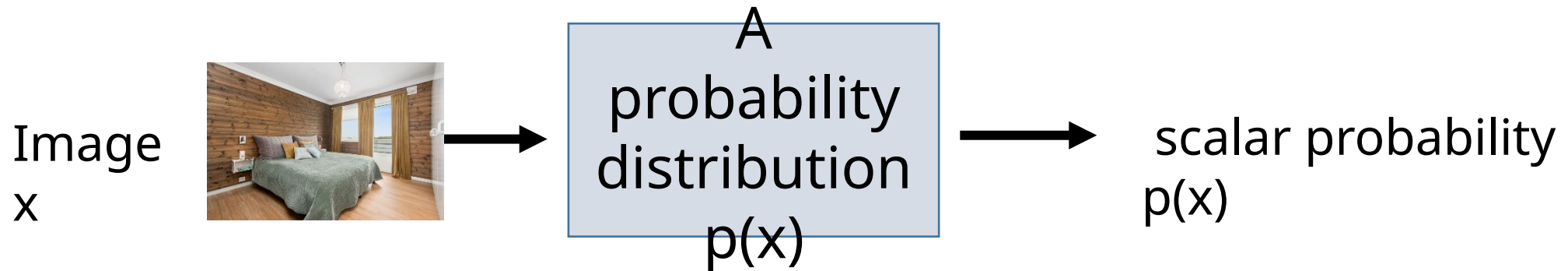
Priors are always necessary, but there is a spectrum



Statistical Generative Models

A statistical generative model is a **probability distribution** $p(x)$

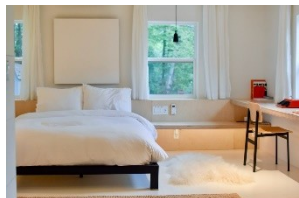
- **Data:** samples (e.g., images of bedrooms)
- **Prior knowledge:** parametric form (e.g., Gaussian?), loss function (e.g., maximum likelihood?), optimization algorithm, etc.



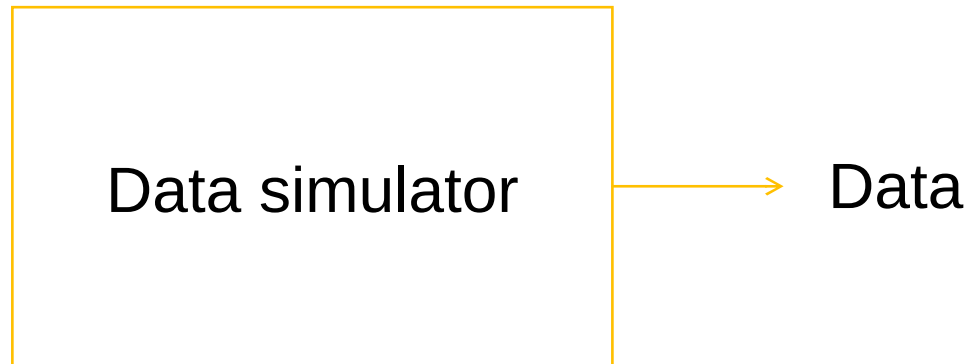
It is generative because **sampling from $p(x)$ generates new images**



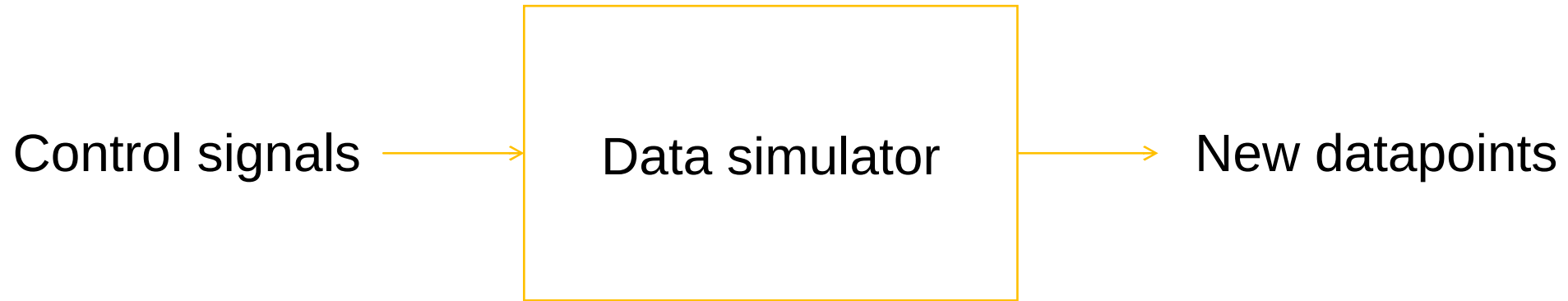
...



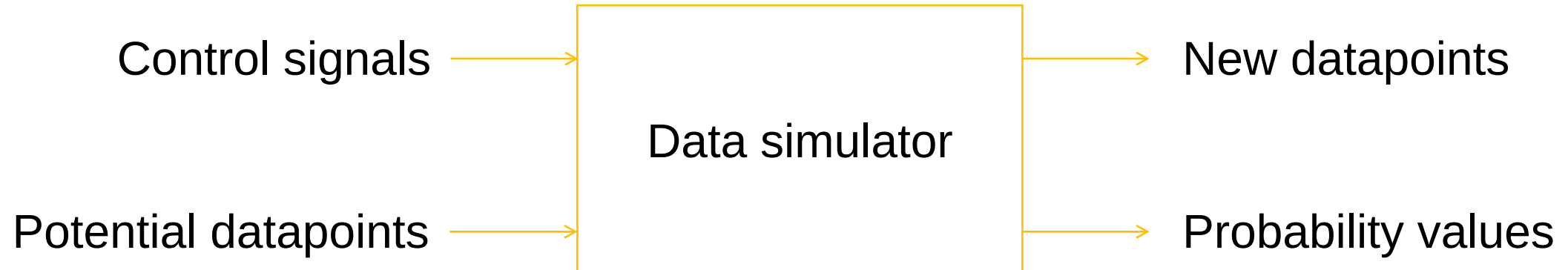
Building a simulator for the data generating process



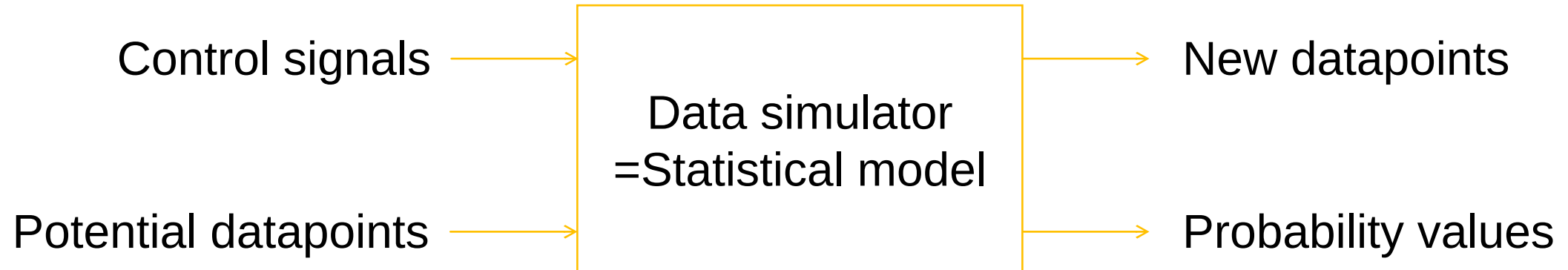
Building a simulator for the data generating process



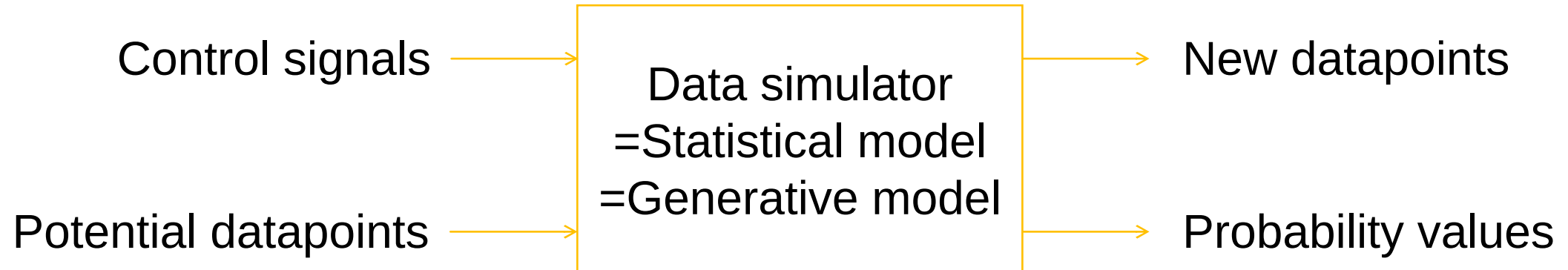
Building a simulator for the data generating process



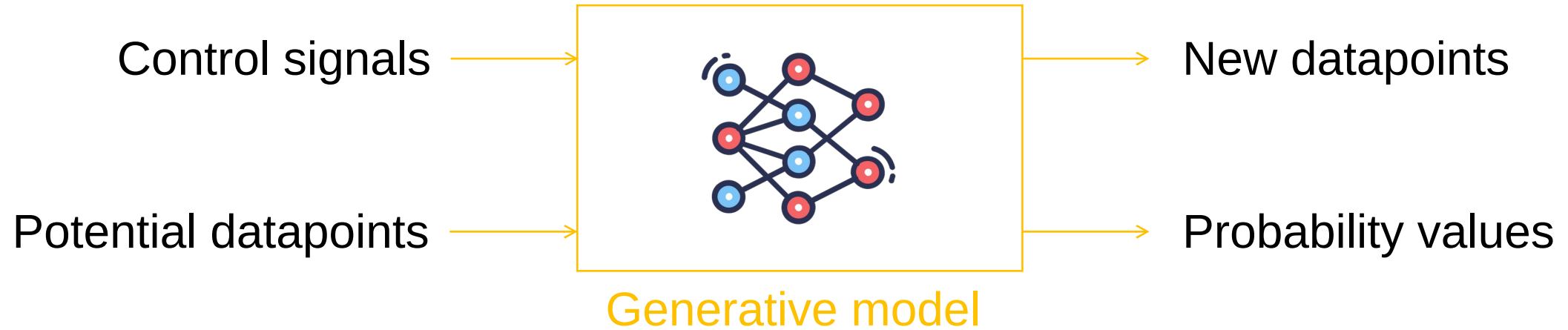
Building a simulator for the data generating process



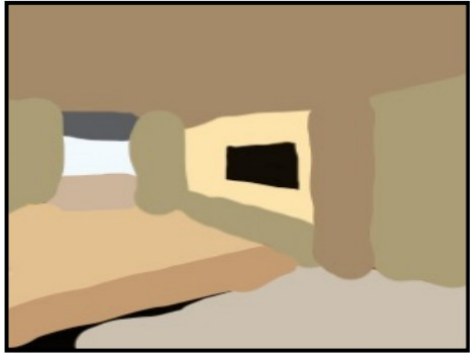
Building a simulator for the data generating process



Building a simulator for the data generating process



Data generation in the real world



Generative model
of realistic images



Stroke paintings to realistic images

[Meng, He, Song, et al., ICLR 2022]

“Ace of Pentacles”



Generative model
of paintings



Language-guided artwork creation

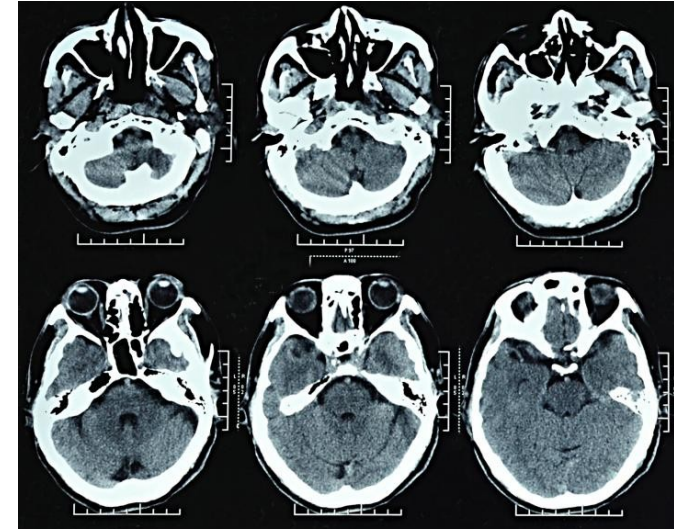
<https://chainbreakers.kath.io> @RiversHaveWings

Solving inverse problems with generative models



Generative model
of medical images

Generate



Medical image reconstruction

[Song et al., ICLR 2022]

Outlier detection with generative models



High
probability
→



Generative model
of traffic signs

←
Low
probability



Outlier detection
[Song et al., ICLR 2018]

Progress in Generative Models of Images -- GANs



2014



2015



2016



2017



2018

Ian Goodfellow,
2019

Progress in Generative Models of Images – Diffusion Models



Text2Image Diffusion Models

User input:

An astronaut riding a horse



Text2Image Diffusion Models

User input:

A perfect Italian
meal



Text2Image Diffusion Models

User input:

泰迪熊穿着戏服，站在太和殿前唱京剧

A teddy bear, wearing a costume, is standing in front of the Hall of Supreme Harmony and singing Beijing opera



Dalle3

A minimap diorama of a cafe adorned with indoor plants. Wooden beams crisscross above, and a cold brew station stands out with tiny bottles and glasses



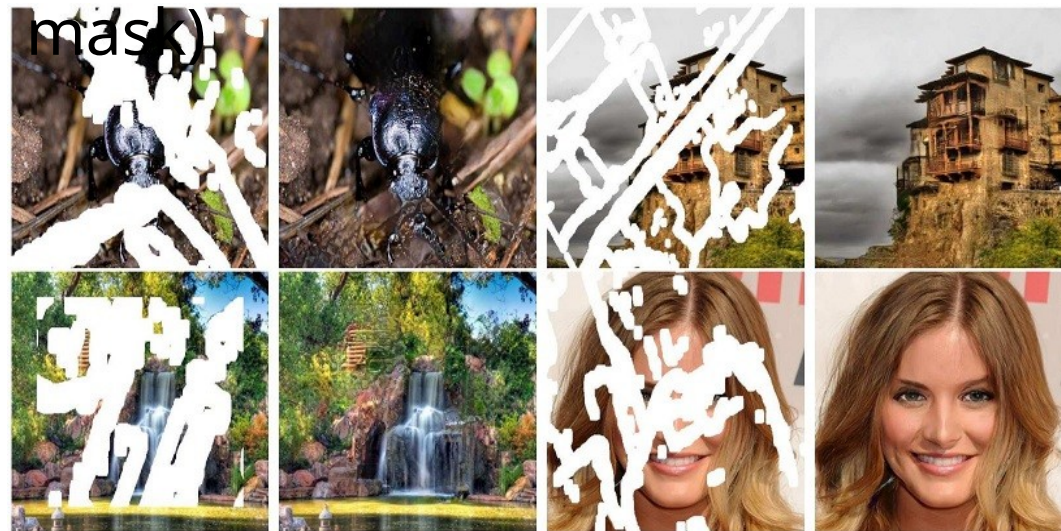
Progress in Inverse Problems

$P(\text{high resolution} \mid \text{low resolution})$



Menon et al, 2020

$P(\text{full image} \mid \text{mask})$



Liu al, 2018

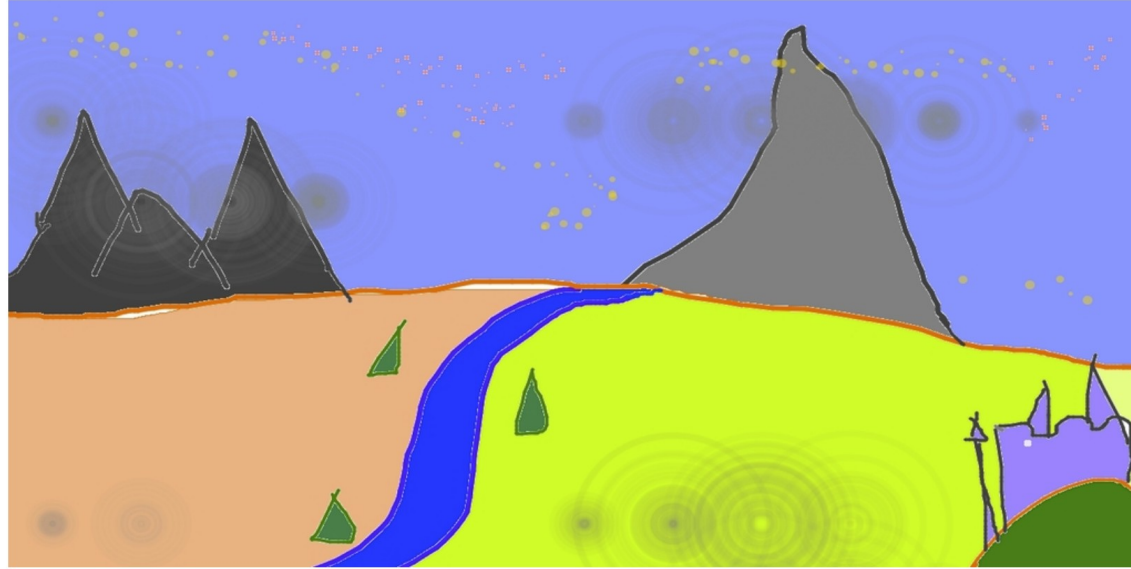
$P(\text{color image} \mid \text{greyscale})$



Antic, 2020

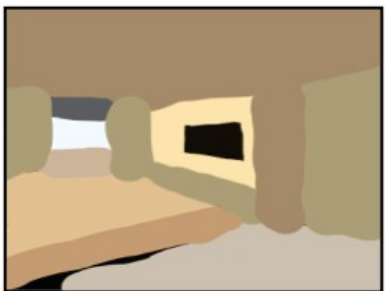
Progress in Inverse Problems

User input:



Progress in Inverse Problems

Stroke Painting to Image



Input

Output

Stroke-based Editing

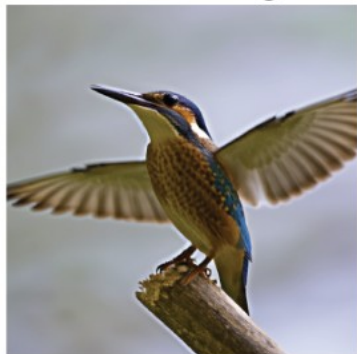


Progress in Inverse Problems

Input Image



Edited Image



“A bird spreading wings”

Input Image



Edited Image



“Two kissing parrots”

Input Image



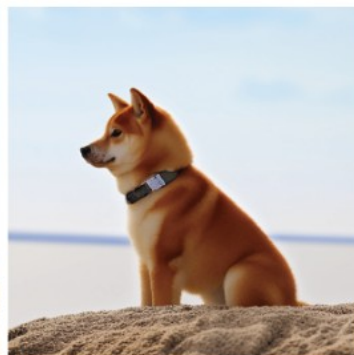
Edited Image



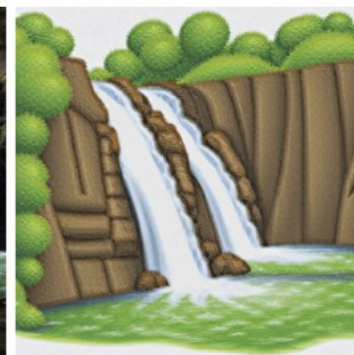
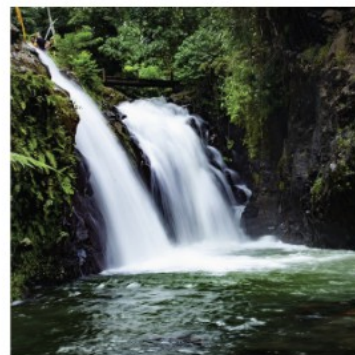
“A goat jumping over a cat”



“A photo of an open box”

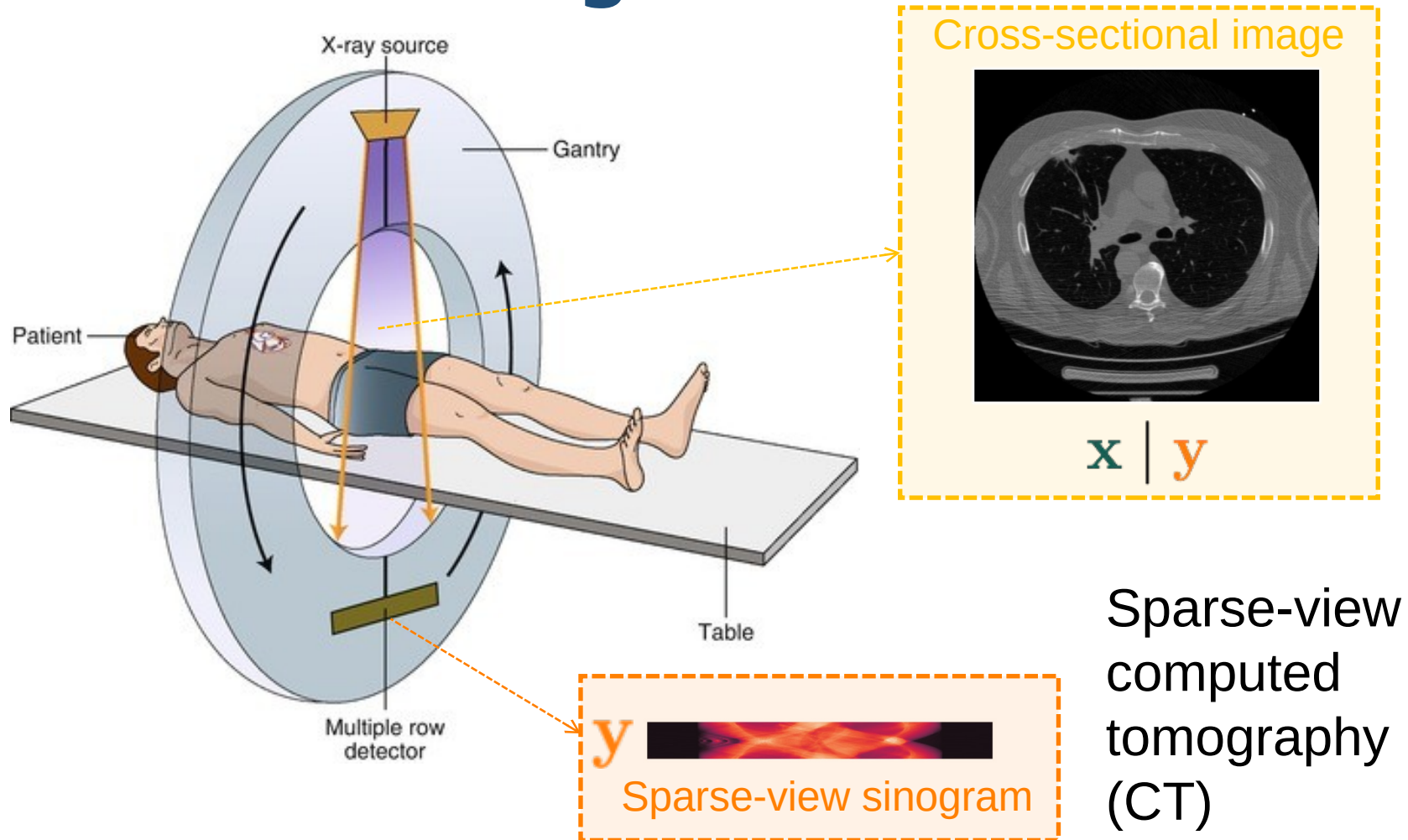


“A photo of a sitting dog”



“A children’s drawing of a waterfall”

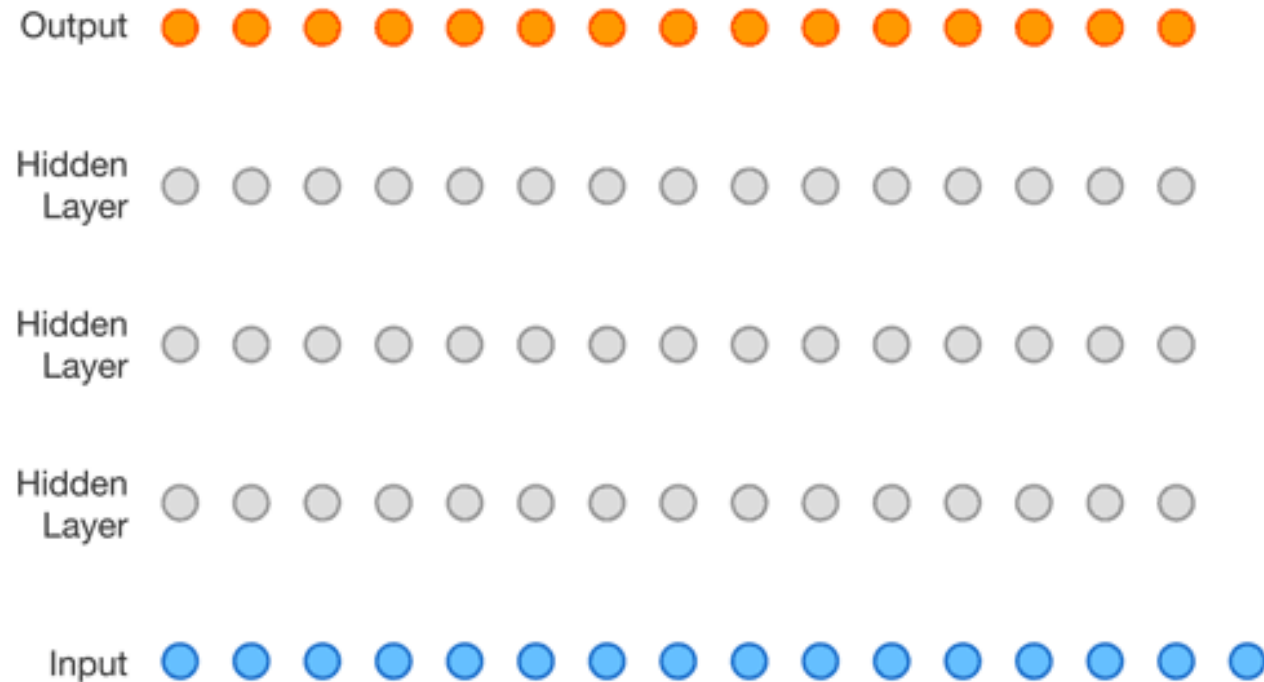
Medical image reconstruction



Forward model $p(\mathbf{y} \mid \mathbf{x})$ is given by physical simulation

WaveNet

Generative model of speech signals



Text to
Speech

Parametr
ic

Concatenati
ve

WaveN
et

Uncondition
al

Musi
c

van den Oord et al,
2016c

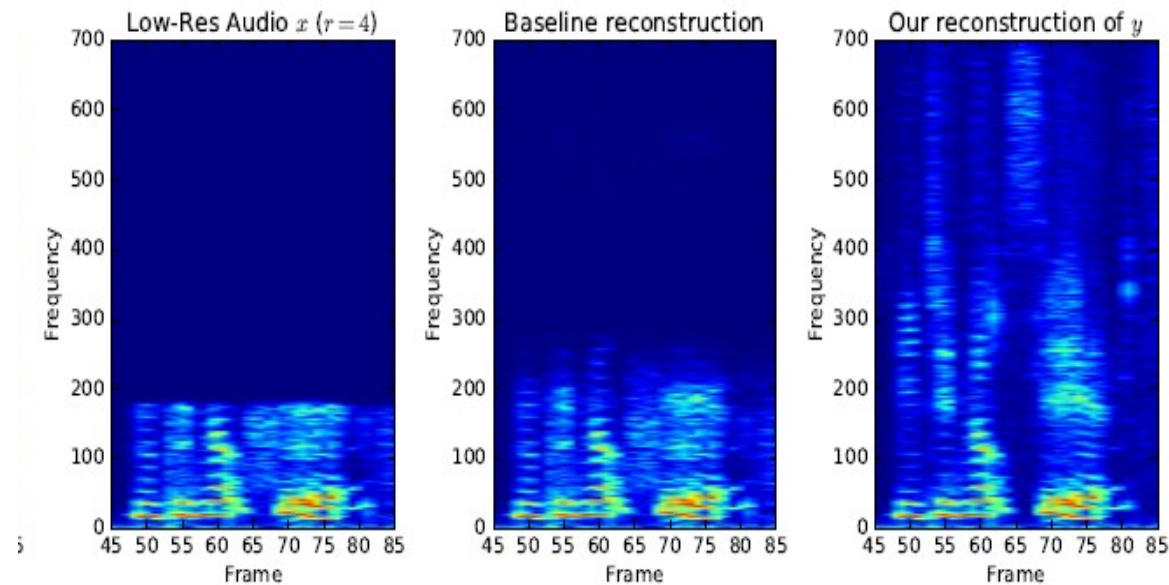
Diffusion Text2Speech

Generative model of speech signals



Audio Super Resolution

Conditional generative model $P(\text{high-res signal} \mid \text{low-res audio signal})$



Low res
signal

High res audio
signal

Kuleshov et al.,
2017

Language Generation

Custom prompt

To get an A+ in deep generative models, students have to

Completion

To get an A+ in deep generative models, students have to be willing to work with problems that are a whole lot more interesting than, say, the ones that most students work on in class. If you're a great student, the question above can be avoided and you'll be able to do great work, but if you're not, you will need to go beyond the basics before getting good.

Now to be clear, this advice is not just for the deep-learning crowd; it is good advice for any student who is taking his or her first course in machine learning.

The key point is that if you have a deep, deep brain of a computer scientist, that's just as important to you.

P(next word | previous words)

Radford et al., 2019
Demo from
talktotransformer.com

Language Generation -- ChatGPT

ER

What should I do to get an A+ in CS236 at Stanford?

To get an A+ in CS236 (Deep Generative Models) at Stanford, you will need to excel in both your understanding of the material and your performance in assignments and exams. Here are some general tips to help you achieve this:

1. ****Attend Lectures and Engage Actively****: Attend all lectures and actively engage with the material. Take thorough notes, ask questions, and participate in discussions. This will help you understand the concepts better.
2. ****Read the Assigned Material****: Make sure to read the assigned textbooks, papers, and supplementary materials. Understanding the theoretical foundations is crucial.
3. ****Stay Organized****: Keep a well-organized notebook or digital notes. This will help you quickly review and understand the material.
4. ****Seek Help When Needed****: Don't hesitate to ask questions if you're having trouble with a concept. You can ask the professor, teaching assistants, or classmates for clarification.
5. ****Complete Assignments Thoroughly****: Take your time to complete assignments, ensuring you fully understand the requirements and concepts involved. Start early and seek help if you're stuck. High-quality assignments are often a significant portion of your grade.
6. ****Collaborate, but Don't Plagiarize****: Collaboration is often encouraged, but make sure you understand your institution's policy on collaboration and plagiarism. Always give credit where it's due and submit original work.
- ...
15. ****Meet with the Professor****: If you're aiming for an A+, consider scheduling meetings with the professor to discuss your progress and seek feedback.

Remember that getting an A+ can be highly competitive, and the specific grading criteria may vary from one course to another and one professor to another. It's essential to understand the grading policies and expectations of your instructor. Always aim for excellence, but also keep in mind that learning and understanding the material should be your primary goal. Good luck!

Machine Translation

Conditional generative model $P(\text{English text} | \text{Chinese text})$

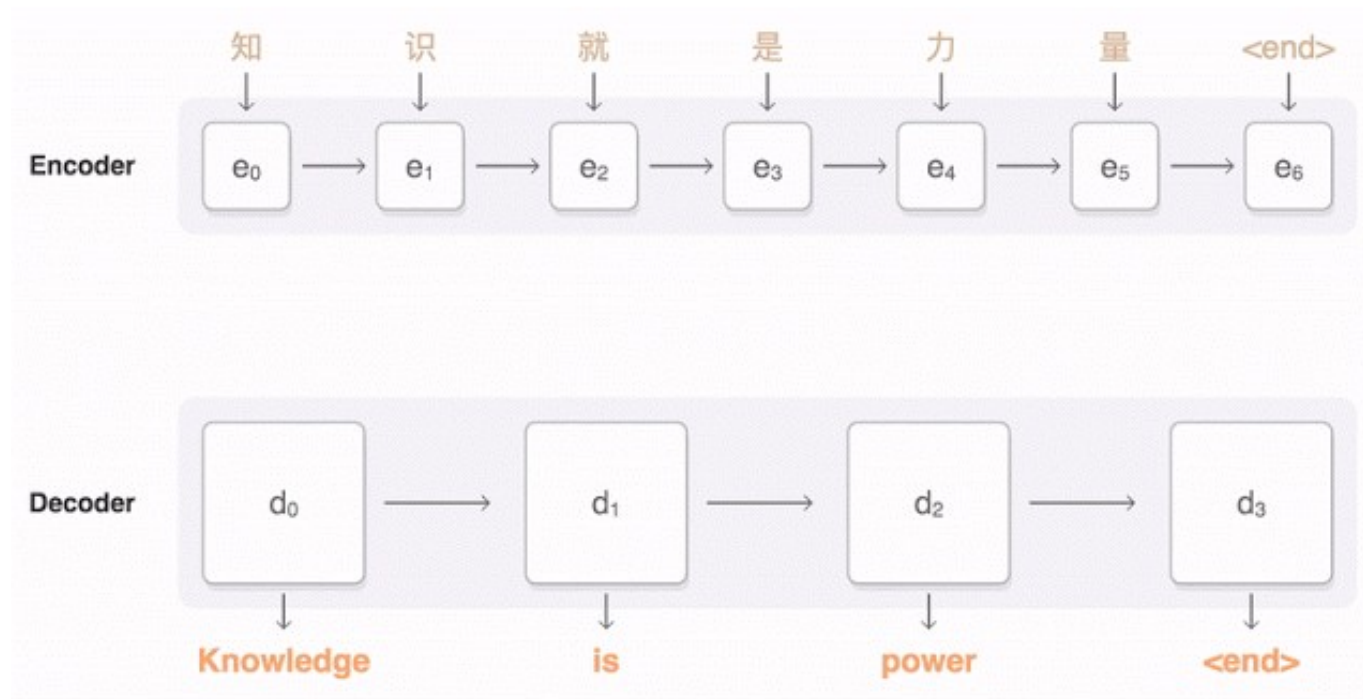


Figure from Google AI research blog.

Code Generation

🐍 parse_expenses.py

🚀 write_sql.go

TS sentiment.ts

📄 addresses.rb

```
1 import datetime
2
3 def parse_expenses(expenses_string):
4     """Parse the list of expenses and return the list of triples (date, va
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
```

Video Generation

Suddenly, the walls of the
embankment broke and there was a
huge flood



Video Generation

a couple sledding down a snowy hill
on a tire roman chariot style



Video Generation

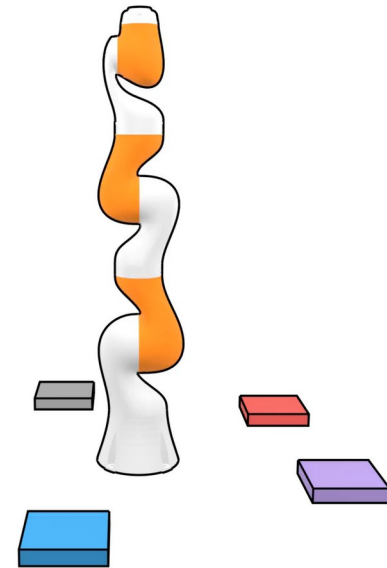


Imitation Learning

Conditional generative model $P(\text{actions} \mid \text{past observations})$

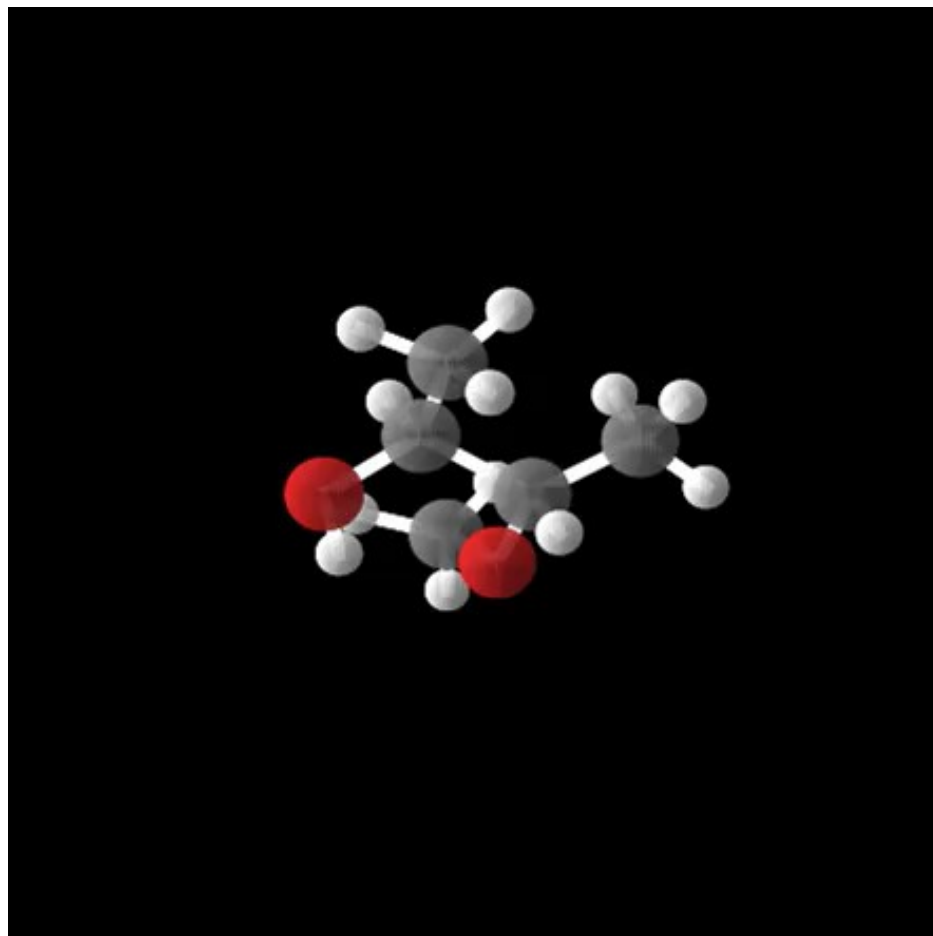


Li et al., 2017



Janner et al., 2022

Molecule generation

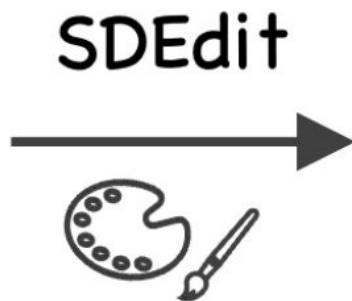


DeepFakes

Which image is real?

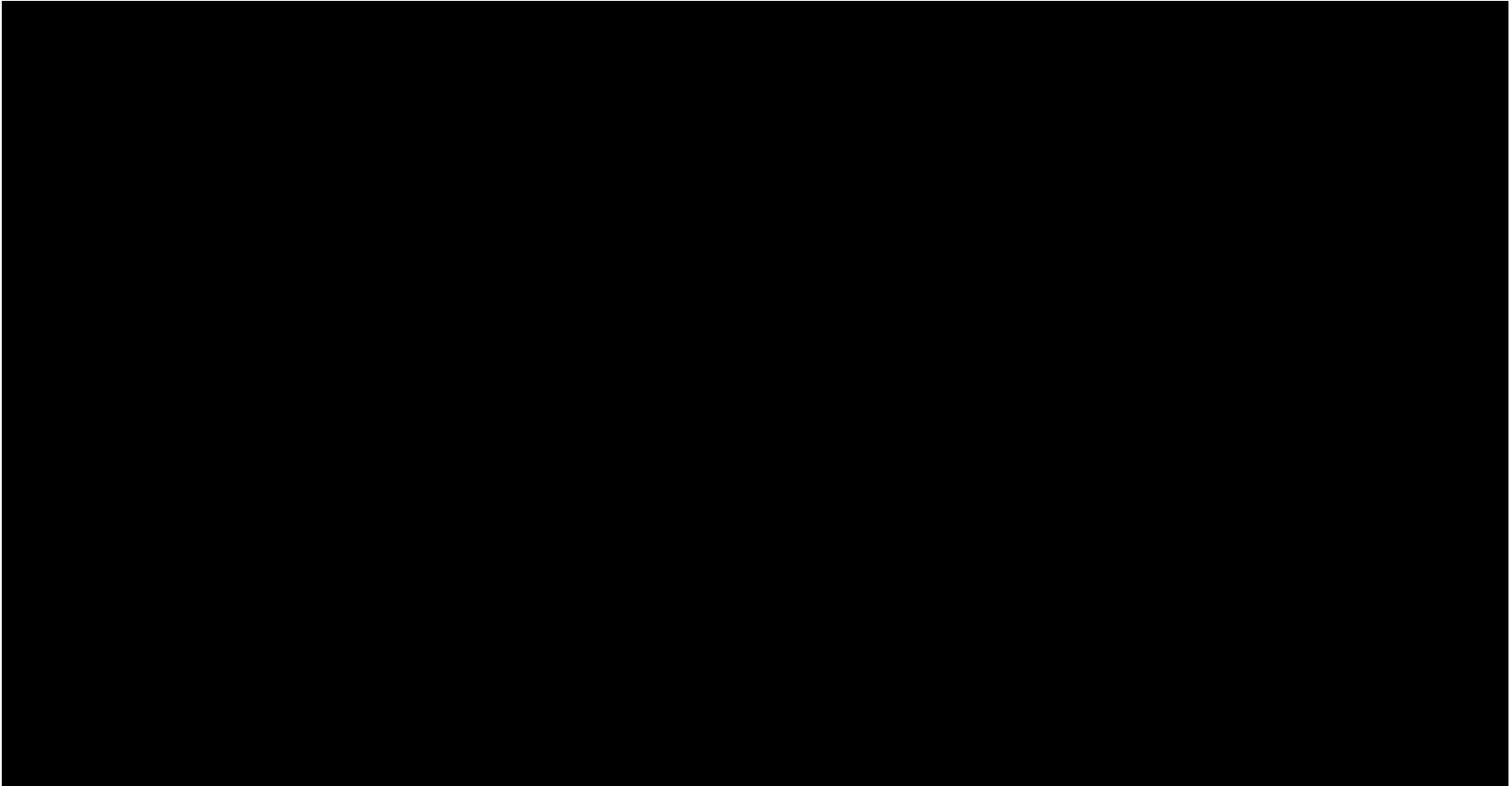


User
 @StefanoErmon



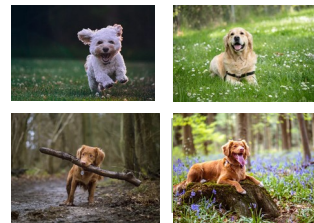
Output

DeepFakes

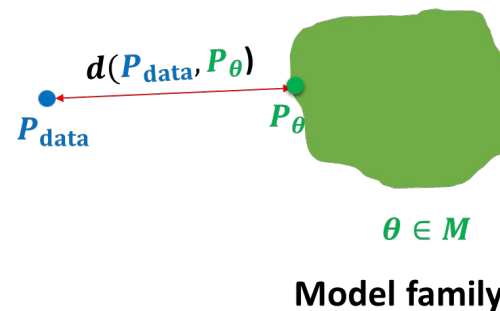


Roadmap and Key Challenges

- **Representation:** how do we model the joint distribution of many random variables?
 - Need compact representation
- **Learning:** what is the right way to compare probability distributions?



$x_i \sim P_{\text{data}}$
 $i = 1, 2, \dots, n$



- **Inference:** how do we invert the generation process (e.g., vision as inverse graphics)?
 - Unsupervised learning: recover high-level descriptions (features) from raw data

Syllabus

- Fully observed likelihood-based models
 - Autoregressive
 - Flow-based models
- Latent variable models
 - Variational learning
 - Inference amortization
 - Variational autoencoder
- Implicit generative models
 - Two sample tests, embeddings, F-divergences
 - Generative Adversarial Networks
- Energy Based Models
- Score-based Diffusion Generative Models
- Learn about algorithms, theory & applications

Prerequisites

- Basic knowledge about machine learning from at least one of CS 221, 228, 229, or 230.
- Basic knowledge of probabilities and calculus:
 - Gradients, gradient-descent optimization, backpropagation
 - Random variables, independence, conditional independence
 - Bayes rule, chain rule, change of variables formulas
- Proficiency in some programming language, preferably Python, required.

Logistics

- Class webpage: <https://deepgenerativemodels.github.io/>
- There is no required textbook. Reading materials and course notes will be provided.
- Suggested Reading: *Deep Learning* by Ian Goodfellow, Yoshua Bengio, Aaron Courville. Online version available free [here](#).
- Lecture notes: <https://deepgenerativemodels.github.io/notes/index.html>
- Teaching Assistants:
 - Agarwal, Pratyush
 - Chatterjee, Soumya
 - Chen, Honglin
 - Chiang, Bryan
 - Obbad, Elyas
 - Salah, Kamyar
 - Shih, Boyun (Andy)
 - Xiao, Zedion (Mark)
 - Xu, Minkai
 - Yuan, Lu (Sylvia)
 - Zhou, Lin Qi (Alex)
- Coordinator: John Cho
- Office hours: See calendar on class website

Logistics – Grading policies

- Grading Policy
 - Three homeworks (15% each): mix of conceptual and programming based questions
 - Midterm: 15%
 - Course Project: 40%
 - Proposal: 5%
 - Progress Report: 10%
 - Poster Presentation: 10%
 - Final Report: 15%

Projects

- Course projects will be done in groups of up to 3 students and can fall into one or more of the following categories:
 - Application of deep generative models on a novel task/dataset
 - Algorithmic improvements into the evaluation, learning and/or inference of deep generative models
 - Theoretical analysis of any aspect of existing deep generative models
- Teaching staff will suggest possible projects
- We will provide Google Cloud coupons
- See list of past projects at:
<https://deepgenerativemodels.github.io/2019>